

host resistance + tolerance

RESISTANCE

host ability to
minimize infection
(parasite clearance)

RQ (host/parasite)

↓
virulence (parasite)

↓
resist/ tolerance (host)

TOLERANCE

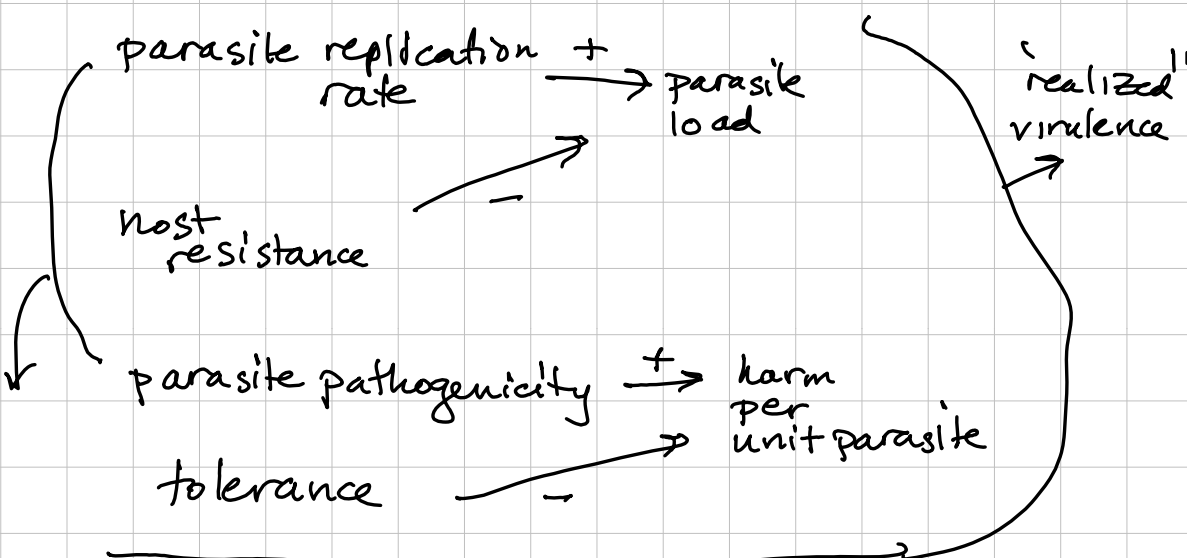
ability to support infection
without losing fitness

RESISTANCE: good for the host, bad for the parasite

TOLERANCE: " " " " , neutral for parasite

VIRULENCE: bad for the host, neutral for parasite
(unless tradeoff)

PATHOGENICITY: harm to host per unit
parasite load.



MECHANISM: host defenses.

CONSTITUTIVE - 'always on'

FACULTATIVE/plastic/adaptive - RESPOND to parasites.

- RECOGNITION systems (RQ)
(eg antibodies)

→ VERTEBRATES: coded by
major histocompatibility complex (MHC)

- Somatic recombination

→ EFFECTORS -

constitutive resistance:

- changing cell surface receptors.

CCR5-Δ32 - HIV resistance.

why aren't all hosts resistant?

- (maybe parasites evolve faster?)
- costs of maintaining recognition systems.
- costs of choosing different habitats
(close ENCOUNTER filter)
- costs of mounting a defense
(energy, immunopathology)

RESISTANCE evolution vs
TOLERANCE evolution

few hosts are resistant

→ don't get infected

→ overall parasite prev. goes down

→ expect RESISTANCE alleles
will equilibrate at some intermediate
frequency when costs + benefits
balance

POLYMORPHISM

→ TOLERANCE:

tolerance might increase parasite spread

higher prevalence → greater fitness
benefit of tolerance

expect FIXATION of tolerance alleles
(→ 100%)